

Clustering of Spatial Data

An application to early childhood education and care services
and their impact on women's labour market

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Introduction

Goal Detect spatial clusters that capture the joint variation in ECEC availability and female employment outcomes in Italian provinces

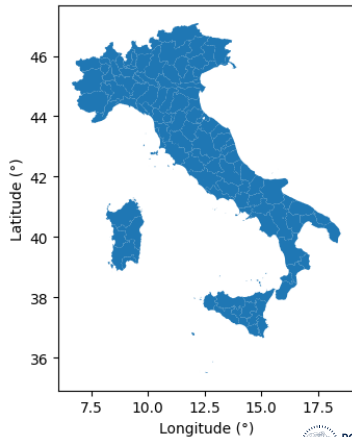
Steps to reach the goal

- Data cleaning and Exploratory Data Analysis
- Variable Selection
- Bayesian CAR model
- Extension to panel data

Dataset Overview

Analysis of a geojson file

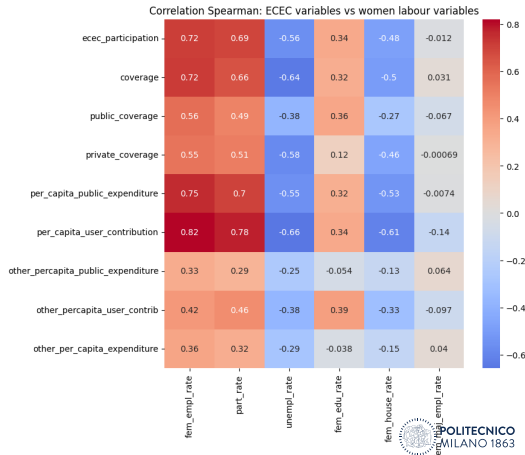
- Data from 2022
- Main source: ISTAT
- 107 provinces
- 47 numerical variables
16 non numerical variables
- No missing values



Exploratory Data Analysis - Correlation

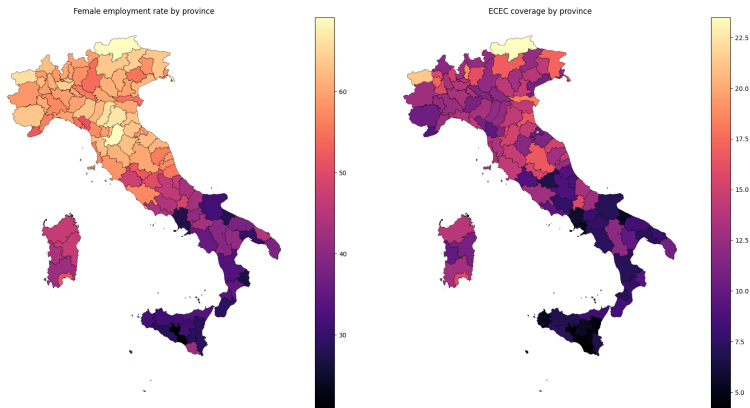
Analysis of the numerical variables correlation

- Subset of relevant variables related to women labour and ECEC services
- Find which variables are strongly correlated, especially to the female employment rate:
 1. per_capita_user_contribution
 2. per_capita_public_expenditure
 3. ecec_partecipazione + coverage



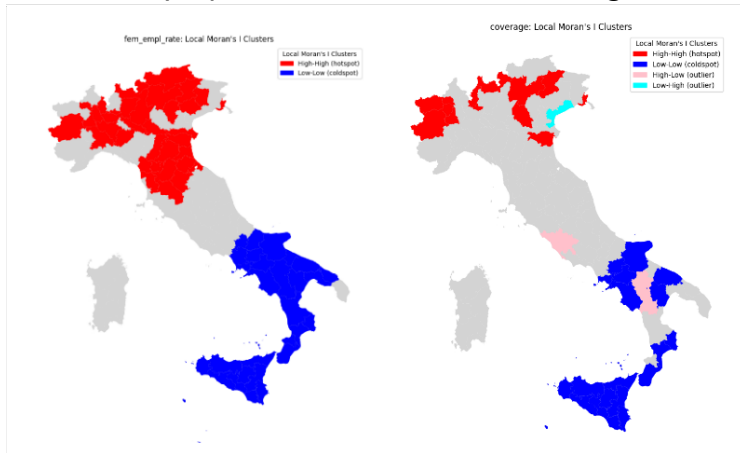
Exploratory Data Analysis - Spatial Visualization

Female employment rate and ECEC coverage by province



Exploratory Data Analysis - Moran's I Cluster

Female employment rate and ECEC coverage cluster



Variable Selection

Removed before selection

- Categorical covariates
- Covariates linked to employment aspects (such as `empl_rate`, `part_rate`, etc.)

First selection

- Remove redundant predictors that are very highly correlated:

$$|\rho_{jk}| \geq 0.90$$

Variable Selection

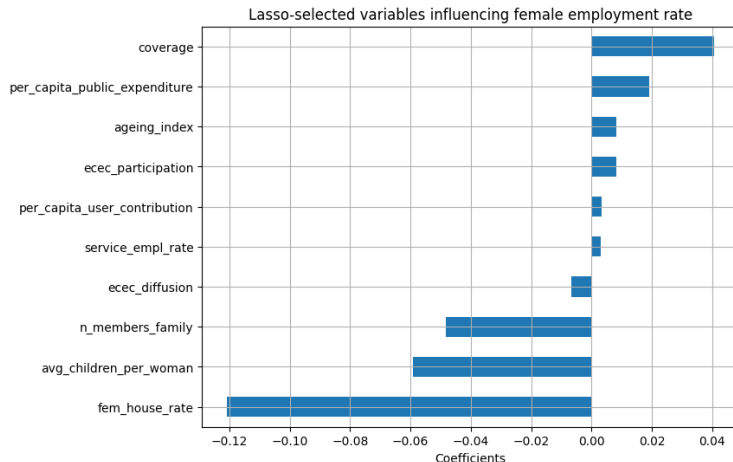
Preprocessing

- Start from the reduced set of covariates $X \in \mathbb{R}^{n \times p}$, after correlation-based filtering
- Standardized predictors and link function:

$$X_{ij}^* = \frac{X_{ij} - \bar{X}_j}{s_{X_j}}, \quad y_i^* = \text{logit}(y_i) = \log\left(\frac{y_i}{1 - y_i}\right)$$

- Define an offset $o_i = \text{logit}(\text{mal_empl_rate}_i)$ to add to the regression model, so that it explains female employment relative to male employment.

Variable Selection - Frequentist Lasso



Model :

$$y_i^* = \alpha + x_i^{*T} \beta + o_i$$

for $i = 1, \dots, n$

10 selected variables
(5 ECEC covariates)

$$R^2 : 0.553 \pm 0.169$$

Variable Selection - Bayesian Lasso

- Bayesian Lasso regression model [1]:

$$y_i^* \mid \beta, \sigma \stackrel{\text{iid}}{\sim} \mathcal{N}(\alpha + x_i^{*\top} \beta + o_i, \sigma^2), \quad i = 1, \dots, n,$$

$$\beta_j \mid \lambda, \sigma \stackrel{\text{iid}}{\sim} \text{Laplace}(0, \sigma/\lambda), \quad j = 1, \dots, p$$

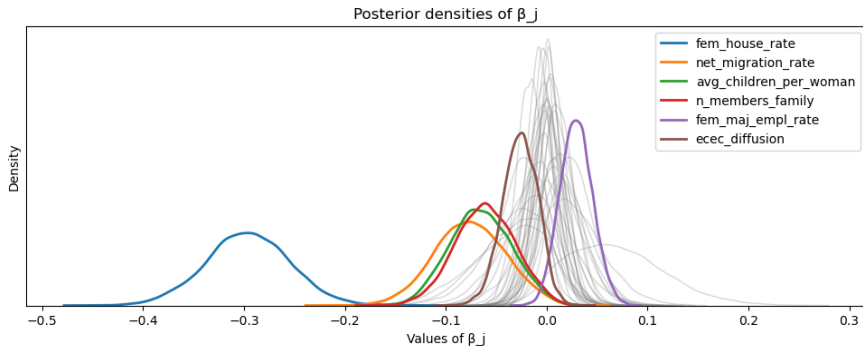
$$\lambda^2 \sim \text{Gamma}(2, 1)$$

$$\sigma \sim \text{HalfNormal}(1)$$

$$\alpha \sim \mathcal{N}(0, 10^2)$$

- Posterior sampling via MCMC (PyMC) for all β_j
- Variable selection rule:
 - Compute 90% credible intervals for each β_j
 - Keep covariates whose 90% credible interval does not contain 0

Variable Selection - Bayesian Lasso Results



Selected Variables: fem_house_rate, net_migration_rate, avg_children_per_woman, n_members_family, fem_maj_empl_rate, ecec_diffusion.

Variable Selection - Comparison

Variable selected	Frequentist Lasso	Bayesian Lasso
fem_house_rate	X	X
net_migration_rate		X
avg_children_per_woman	X	X
n_members_family	X	X
fem_maj_empl_rate		X
ecec_diffusion	X	X
per_capita_public_expenditure	X	
ecec_participation	X	
coverage	X	
per_capita_user_contribution	X	
ageing_index	X	
service_empl_rate	X	



CAR Model

Model Formulation with CAR Leroux prior:

$$y_i^* | \alpha, x_i^*, \beta, o_i \stackrel{\text{ind}}{\sim} \mathcal{N}(\alpha + x_i^{*\top} \beta + \phi_i + o_i, \sigma^2), \quad i = 1, \dots, n$$

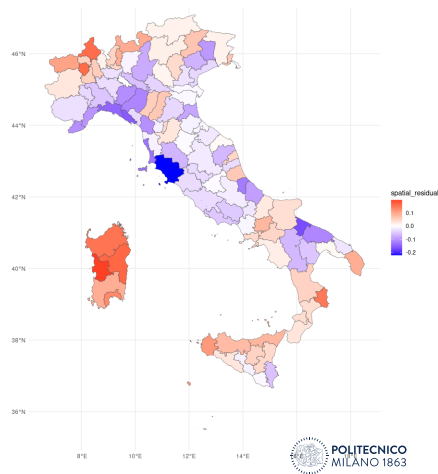
$$\alpha \sim \mathcal{N}(0, 1000)$$

$$\beta \sim \mathcal{N}(0, 1000 \mathbf{I}_p)$$

$$\phi_i | \phi_{-i}, W, \tau^2, \rho \stackrel{\text{ind}}{\sim} \mathcal{N}\left(\frac{\rho \sum_{j=1}^n w_{ij} \phi_j}{\rho \sum_{j=1}^n w_{ij} + 1 - \rho}, \frac{\tau^2}{\rho \sum_{j=1}^n w_{ij} + 1 - \rho}\right)$$

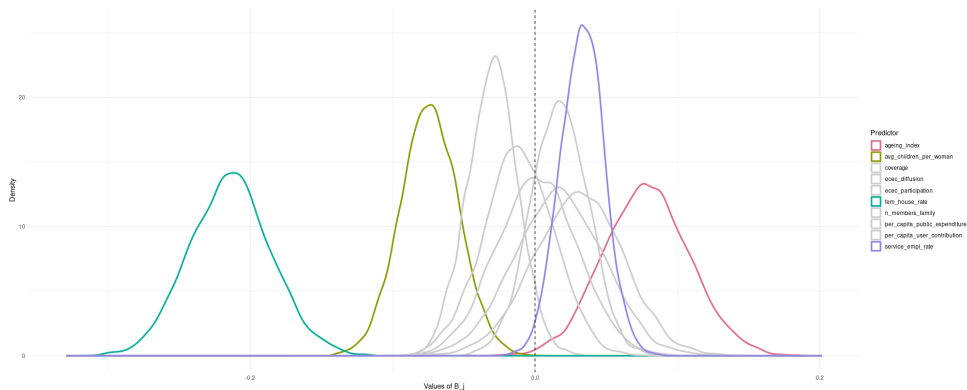
$$\tau^2 \sim \text{inv-gamma}(1, 0.01)$$

$$\rho \sim \mathcal{U}(0, 1)$$



Posterior Mean of CAR spatial effect

CAR model



Posterior densities of β_j . The colored densities belong to covariates whose coefficients' 95% CI does not include zero. Model fitted through CARBayes [2]

Future Steps

Idea for the final model

- Investigate the differences between frequentist and bayesian variable selection
- Model our target variable as a mixture of distributions
- Improve areal random effect with spatial association (Conditional AutoRegressive (CAR) PRIOR): provinces that are close to each other tend to share similar characteristics
- Use a Dirichlet Process Prior on the coefficients. Instead of assuming that all the provinces share the same relationships with the covariates, this approach lets data group provinces into clusters that have different patterns

References

- [1] Park, T. and Casella, G. (2008) "The Bayesian Lasso." Journal of the American Statistical Association, 103, 681-686.
- [2] Lee, D. (2013). "CARBayes: An R Package for Bayesian Spatial Modeling with Conditional Autoregressive Priors." Journal of Statistical Software, 55(13), 1-24.

Appendix - Variables Details I

Highlighted rows show computed variables, not directly taken.

Variable	Description
fem_empl_rate	Employed women (15–64) / total women (15–64)
population	Residents on January 1st
unempl_rate	Unemployed (15–64) / labor force (15–64)
part_rate	Labor force (15–89) / population (15–89)
empl_rate	Employed (15–64) / labor force (15–64)
empl_growth	Annual growth of employed population
inact_rate	Inactive (15–64) / population (15–64)
fem_edu_rate	Women 20+ with tertiary education
male_edu_rate	Men 20+ with tertiary education
service_empl_rate	Workers in services (15–64) / all workers
GDP	Gross Domestic Product at market prices per person
n_large_comp	Local units with 250+ employees / active population (in thousand)
n_self_empl	Number of self-employed / active population
young_neet	15–29 not in employment, education or training / 15–29 people

Appendix - Variables Details II

Variable	Description
fem_house_rate	Inactive women / children aged 0–2
foreign_rate	Foreigners / total population
n_members_family	Average number of members per family
elderly_in_care	65+ in home or integrated care services
house_price	Average price of housing (€/sqm)
fertility_rate	Live births per 1,000 residents
first_birth_maternal_age	Average age at first childbirth
avg_children_per_woman	Total fertility rate (TFR)
dependency_rate	Non-working / working-age population
ageing_index	65+ / 0–14 population
net_migration_rate	Net migration per 1,000 residents
adj_net_migration_rate	Migration adjusted for natural change
area_km2	Total surface area

Appendix - Variables Details III

Variable	Description
public_transport_supply	Public transport supply per capita
internet_coverage_rate	% of households with high-speed internet
graduate_mobility_rate	Net migration of graduates (25–39)
green_m2_capita	Green space per capita in provincial capitals
eco_urban_index	Environmental performance score
per_capita_public_expenditure	Municipal spending per 0–2 year-old
eccec_participation	% of 0–2 y.o. using early childhood services
coverage*	Number of available childcare services / children aged 0-2
urban_type	1: Urban, 2: Intermediate, 3: Rural
per_capita_user_contribution	Per-child user spending (€ per 0–2 y.o.)
advanced_business_empl	Employees in advanced business services per active population
lifelong_learn	% of population aged 25–64 in lifelong learning
startup_ratio	Innovative startups per 1,000 corporations
fem_maj_empl_rate	Workers in Education and Health and Social Care / all workers